CONFIDENTIAL **MOCK EXAM 2026**

GENERIC UNIVERSITY FINAL EXAMINATION

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

PART A (80 MARKS)

QUESTION 1

a) Using Caesar Cipher technique, decrypt the following ciphertext with a shift of 7 (replace each letter with the one 7 positions to the left).
KLCPJLYLABYULKHUKDPWLKVUL

(2 marks)

b) State six facts about substitution ciphers as presented in the lecture, covering what the cipher does to plain-text letters, what serves as the key, the resulting output, the Shannon principle it corresponds to, and where the idea is applied today.

(6 marks)

QUESTION 2

a) List two facts about transposition ciphers as described in the lecture, covering how the rail fence cipher arranges plain-text and what serves as its key.

(2 marks)

b) Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

(6 marks)

Vigenere tableau (A = 0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

QUESTION 3

a) List the stages that make up the identification stage and the authentication stage of a biometric authentication process.

(2 marks)

b) Refer to your answer in (a). Based on the scenario in (a), discuss the biometric mechanisms and technologies used for authentication. Your discussion should cover the definition of a biometric, the two categories of biometric applications, and specific technologies with their characteristics.

(6 marks)

QUESTION 4

a) Explain the Chinese Wall model as a formal security model, covering the policy it implements and how its access rules behave over time.

(2 marks)

b) [COMPUTED PLACEHOLDER - acm, 6 marks]

(6 marks)

QUESTION 5

a) Explain the difference between morals, ethics and laws as they apply to the conduct of a computing professional.

(2 marks)

b) State six professional conduct requirements that the ACM code places on a computing professional, covering both how they carry out their work and how they treat the public and the systems they build.

(6 marks)

QUESTION 6

a) State TWO (2) typical locations where IDS sensors are placed in a network-based IDS architecture.

(2 marks)

b) Describe six (6) preventive network security mechanisms and the protection each provides, as covered in the network security lecture.

(6 marks)

QUESTION 7

a) State TWO (2) points about software insecurity and buffer overflows as described in the Software Security lecture.

(2 marks)

b) [THEORY PLACEHOLDER - software/OS security, 6 marks, Explain. No sourced question passed verification.]

(6 marks)

QUESTION 8

a) [THEORY PLACEHOLDER - access control models, 2 marks, State. No sourced question passed verification.]

(2 marks)

b) Discuss the goals of access control and the role of the reference monitor in authorization, and explain how the Chinese Wall model's simple security rule and *-property constrain access over time. Provide six distinct points.

(6 marks)

QUESTION 9

a) Explain two password security controls that can be applied to password files and accounts to strengthen authentication.

(2 marks)

b) Explain how single sign-on and Kerberos-based authentication mechanisms reduce the burden of password management for users in a large distributed environment, and describe the role of smart cards as an alternative authentication mechanism.

(6 marks)

QUESTION 10

a) Explain two characteristics of Trojan horses as a type of malware, including how they present themselves and what harm they cause once active.

(2 marks)

b) Explain six techniques by which malware such as viruses and worms propagate themselves or attach their code to a host program, and describe how early malware demonstrated these propagation methods.

(6 marks)

PART B (20 MARKS)

QUESTION 1

a) Explain five ways in which threat classes and malware can impact the confidentiality, integrity and availability of an information system, giving a distinct impact for each.

(5 marks)

b) The Samy worm spread through MySpace by exploiting a cross-site scripting flaw. Explain five aspects of this case: how the malicious script was injected, how the worm gathered and added friends, how infection spread to visitors, how the script reached the browser, and the eventual scale of the outbreak.

(5 marks)

QUESTION 2

A hospital wants every workstation to prove at boot that it is running only approved software before it is allowed to access patient records. The IT team is told to base the design on the Trusted Platform Module and a chain of trust, and to explain the components involved.

a) State the role of the Trusted Platform Module in a trusted platform, and explain how a chain of trust is built from the hardware up to the running software, naming the stages involved and the roots of trust that support it.

(5 marks)

b) Referring to the scenario, discuss the NGSCB/Nexus components that the IT team could use to enforce the approved-software requirement, covering the role of Nexus, the role of the Nexus Computing Agent, and how these components relate to the TPM.

(5 marks)

END OF QUESTION PAPER